

Q1. Given a sorted array (distinct elements) count all pairs (i, j) such that $arr[i] + arr[j] = k$. ($i \neq j$)

ex $\Rightarrow$

	0	1	2	3	4	5	6	7	8	9
	-3	0	4	3	6	8	11	14	18	25

k = 17

O/p $\Rightarrow$ Count = 2

Basic $\Rightarrow$ if consider all pairs $\rightarrow O(N^2)$

↓

$$a + b = k$$

↑

$$arr[0] \Rightarrow b = k - arr[0] / a$$

if using Binary search $\rightarrow O(N \log N)$

↓

if use extra space (Hashmap) $\rightarrow TC \Rightarrow O(N)$
 $SC \Rightarrow O(N)$

↓

Two pointers

$$k=17$$

0	1	2	3	4	5	6	7	8	9
-3	0	4	3	6	8	11	14	18	25

i)

i j

j i

ii)

iii)

i j

$$25 + (-3) = 22 > 17$$

$$\downarrow$$

$$25 + 4 > -3 \Rightarrow 22$$

iv)

i

j

i)

$$\text{arr}[i] + \text{arr}[j] = -3 + 0 = -3 < 17$$

ii)

$$\text{arr}[i] + \text{arr}[j] = 18 + 25 = 43 > 17$$

iii)

$$\text{arr}[i] + \text{arr}[j] = 6 + 8 = 14 < 17$$

iv)

$$\text{arr}[i] + \text{arr}[j] = -3 + 25 = 22 > 17$$

i -

$$-3 + 18 = 15 < 17$$

↑

↑

i +

$$0 + 18 = 18 > 17$$

0 1 2 3 4 5 6 7 8 9
-3 0 4 3 6 8 11 14 18 25

i
j

k=17

$$-3 + 25 = 22 > 17 \rightarrow j--$$

$$-3 + 18 = 15 < 17 \rightarrow i++$$

$$0 + 18 = 18 > 17 \rightarrow j--$$

$$0 + 14 = 14 < 17 \rightarrow i++$$

$$1 + 14 = 15 < 17 \rightarrow i++$$

$$3 + 14 = 17 == 17 \rightarrow \text{count}++; i++; j--$$

$$6 + 11 = 17 == 17 \rightarrow \text{" " "}$$

$$i > j$$

return count $\rightarrow$ 2

TC $\Rightarrow O(N)$
SC $\Rightarrow O(1)$

$$i=0; j=N-1; \text{count}=0$$

↓

while (i < j) {

arr[i] + arr[j]

< k
i++

↓
== k
count++
i++, j--

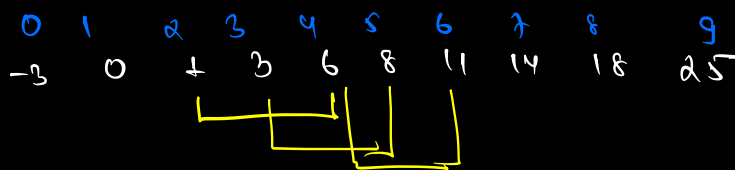
> k
j--

place your pointers properly

make a decision to reject one of ptr. as a possible part of ans.

Q2. Given sorted array (distinct element), count all pairs (i, j) such that $arr[j] - arr[i] = k$. ($i < j$)

$k=5$



count = 3

Basic $\Rightarrow$ all pairs $\rightarrow O(N^2)$

$\downarrow$

binary search $\rightarrow O(N \log N)$

$$A - B = k$$

$\uparrow$

$$\Rightarrow -B = k - A$$

$$\Rightarrow B = \boxed{A - k}$$

$\downarrow$

using hashing $\rightarrow TC \approx O(N) \quad SC \approx O(N)$

↓
two pointers

k=5

0	1	2	3	4	5	6	7	8	9
-3	0	1	3	6	8	11	14	18	25
<u>i</u>									<u>j</u>

$$\text{arr}[j] - \text{arr}[i] = 25 - (-3) = 28 > 5$$

$$A - B \Rightarrow k$$

$$\downarrow \quad \uparrow$$

$$\underline{j--} \quad \underline{i++}$$

0	1	2	3	4	5	6	7	8	9
-3	0	1	3	6	8	11	14	18	25
<u>i</u>			<u>j</u>						

$$\begin{array}{ccc} \text{arr}[j] - \text{arr}[i] & \Rightarrow & 0 - (-3) = 3 < 5 \rightarrow j++ \\ \uparrow & & \downarrow \end{array}$$

$$1 - (-3) = 4 < 5 \rightarrow j++$$

$$\begin{array}{ccc} \text{arr}[j] - \text{arr}[i] & \Rightarrow & 3 - (-3) = 6 > 5 \rightarrow i++ \\ \downarrow & & \uparrow \end{array}$$

$$3 - 0 = 3 < 5$$

$k=5$

	0	1	2	3	4	5	6	7	8	9
	-3	0	4	3	6	8	11	14	18	25

j

$$0 - (-3) = 3 < 5 \rightarrow j++$$

i

$$1 - (-3) = 4 < 5 \rightarrow j++$$

$$3 - (-3) = 6 > 5 \rightarrow i++$$

$$3 - 0 = 3 < 5 \rightarrow j++$$

$$6 - 0 = 6 > 5 \rightarrow i++$$

$$6 - 1 = 5 == 5 \rightarrow \text{curr}++, i++, j++$$

$$8 - 3 = 5 == 5 \rightarrow " , " , "$$

$$11 - 6 = 5 == 5 \rightarrow " , " , "$$

$$14 - 8 = 6 > 5 \rightarrow i++$$

$$14 - 11 = 3 < 5 \rightarrow j++$$

$$18 - 11 = 7 > 5 \rightarrow i++$$

$$18 - 14 = 4 < 5 \rightarrow j++$$

$$25 - 14 = 9 > 5 \rightarrow i++$$

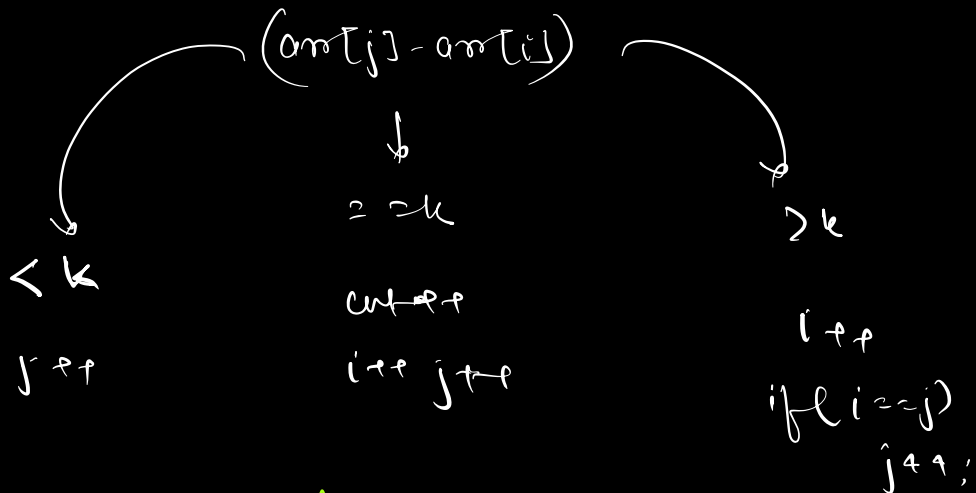
$$25 - 18 = 7 > 5 \rightarrow i++$$

$$\underline{\underline{\text{curr} = 3}}$$

$TC \Rightarrow O(N)$
 $SC \Rightarrow O(1)$

$i=0; j=1; cur=0$

while ($j < N$) {



Q3. Given an array of five integers, find count of subarrays whose sum = k

$k = \underline{\underline{15}}$

3	2	5	1	8	6	2	10
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$O/p \Rightarrow \text{count} = 1$

Brute $\Rightarrow$ all possible subarrays $\Rightarrow O(N^2)$

$\downarrow$
 $O(N^2)$
 $\xrightarrow{\text{conformant}} \underline{\underline{=}}$

arr $\Rightarrow$ 3 2 5 1 8 6 2 10

pf $\Rightarrow$ 3 5 10 11 19 25 27 37

$\downarrow$

sorted
and distinct

$$\begin{aligned} \text{sum of subarr} & \Rightarrow \text{sum}(i-j) \Rightarrow \text{pf}[j] - \text{pf}[i-1] = k \\ & \quad i=0, \quad \text{pf}[j] \end{aligned}$$

$$\boxed{\begin{aligned} \text{TC} &\Rightarrow O(N) \\ \text{SC} &\Rightarrow O(N) \end{aligned}}$$

Q4, Given a sorted array distinct element, find triplets (i, j, k) such that $\text{arr}[i] + \text{arr}[j] + \text{arr}[k] = \text{sum}$

ex $\Rightarrow$ -8 -4 -3 -1 2 3 5 7 9 sum = 14

O/p $\Rightarrow$ count = 2

Basic $\Rightarrow$ consider all possible triplets
 $\rightarrow O(N^3)$

$$A + B + C = \text{sum}$$

↑

$\text{arr}[0]$

$$\Rightarrow B + C = (\text{sum} - \text{arr}[0]) \rightarrow q$$

$$\Rightarrow \boxed{B + C = q}$$

$$\begin{array}{ccccccc} -8 & -4 & [-3 & -1 & 2 & 3 & 5 & 7 & 9] \\ & \downarrow & & & & & & & \\ & \underline{\underline{A}} & & & & & & & \end{array}$$

$$-8 + B + C = 15$$

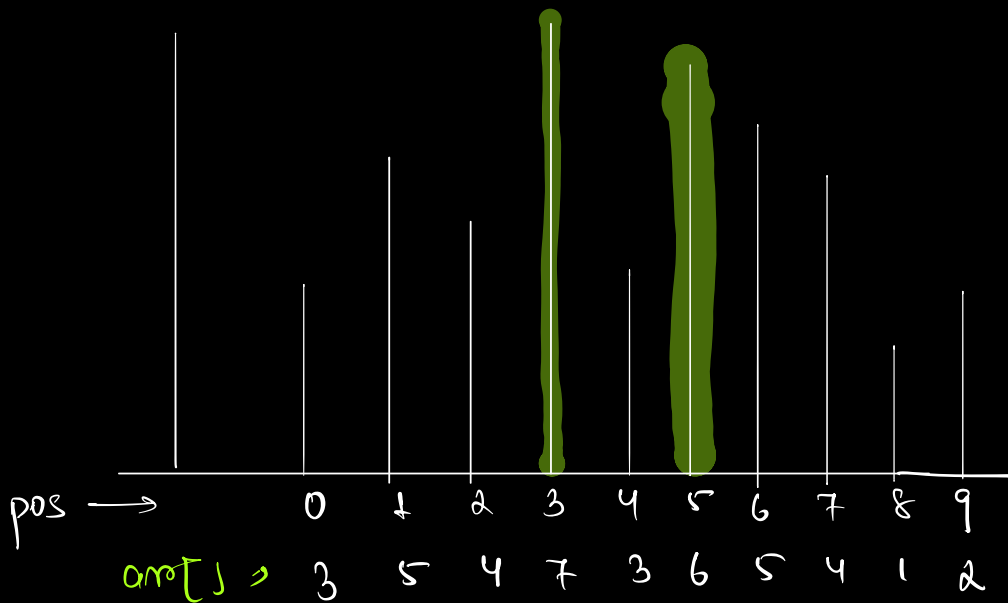
$$\Rightarrow \boxed{B + C = \underline{\underline{23}}} \leftarrow$$

$\{$
 for ($i = 0 \rightarrow N$) $\{$
 for ($\text{arr}[i] \rightarrow A$ $O(N)$
 two pb - ($i+1 \rightarrow N$) $\{$ $\leftarrow$
 solve for $B + C = \text{sum} - A$
 }
 }
 $\}$

$$\boxed{\begin{array}{l} TC \Rightarrow O(N^2) \\ SC \Rightarrow O(1) \end{array}}$$

Q5 Given N arr element $\rightarrow$ heights of walls

find 2 walls, such that they accumulate max water b/w them.



$$(3, 6) \Rightarrow (6-3) \times \min(7, 5)$$

width height

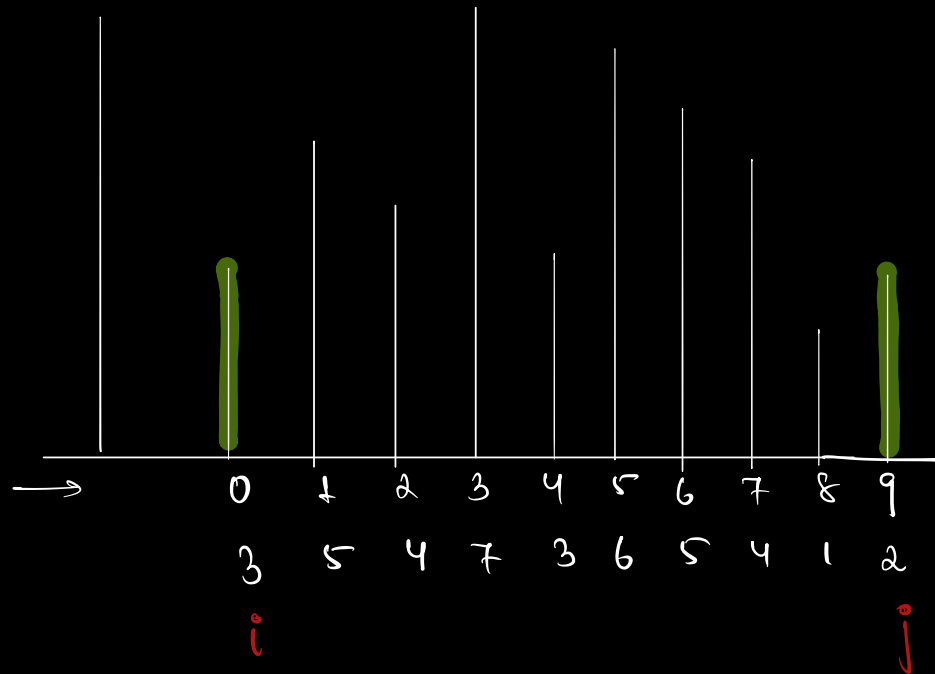
$$\Rightarrow 3 \times 5 = \underline{\underline{15}}$$

$$(3, 5) \Rightarrow (5-3) \times \min(6, 7)$$

$$= 2 \times 6$$

$$= \underline{\underline{12}}$$

$$\text{water}(i, j) = (j - i) \times \min(\text{arr}[i], \text{arr}[j])$$



$$\text{water}(0, 9) = 9 \times \min(2, 3) = 18$$

we have to maximise water = width $\times$ height



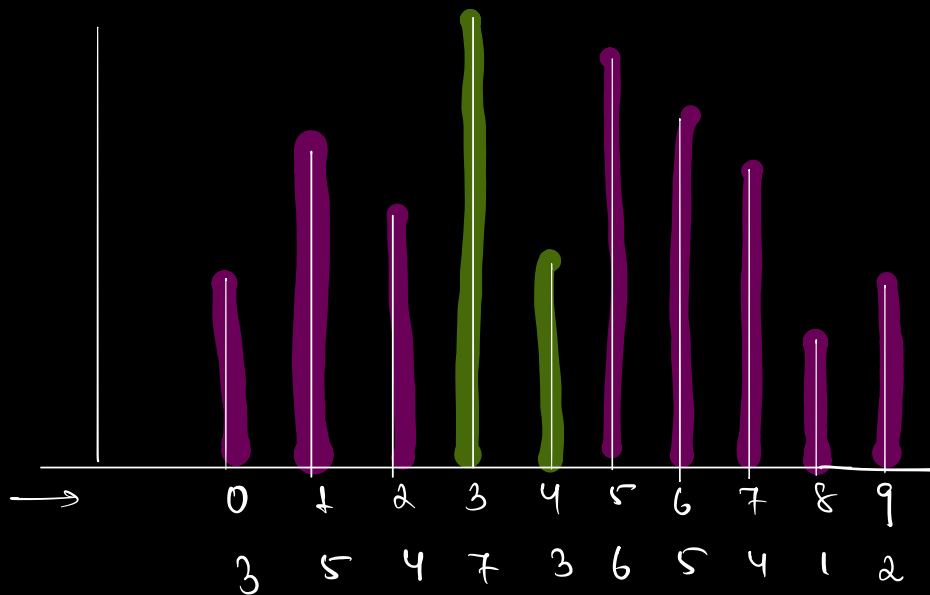
maxim
width is already
achieve



so maximise the
height—



discard smaller height



Ans = 18 ~~24~~ ~~25~~ 25

i
j

$$\text{water}(0,9) = 9 \times \min(2,3) = 18 \rightarrow j--$$

$$\text{water}(0,8) = 8 \times \min(1,3) = 8 \rightarrow j--$$

$$\text{water}(0,7) = 7 \times \min(4,3) = 21 \rightarrow i++$$

$$\text{water}(1,8) = 6 \times \min(4,5) = 24 \rightarrow j--$$

$$\text{water}(1,6) = 5 \times \min(5,5) = 25 \rightarrow i++ \text{ (any)}$$

$$\text{water}(2,6) = 4 \times \min(4,5) = 16 \rightarrow i++$$

$$\text{water}(3,6) = 3 \times \min(7,5) = 15 \rightarrow j--$$

$$\text{water}(3,5) = 2 \times \min(7,6) = 12 \rightarrow j--$$

$$\text{water}(3,4) = 1 \times \min(7,3) = 3 \rightarrow j--$$

$$i = 0, \quad j = N-1$$

```
while ( i < j ) {
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$$ans = \max(ans, (j-i) \times \min(arr[j], arr[i]))$$
$$|b(\text{arr}[i] < \text{arr}[j])$$

٤٢٢

else

j n -

$$\begin{aligned} T_c &= O(N) \\ S_c &= O(1) \end{aligned}$$

⇒ Duplicates

① 1 3 4 5 6 7 10 k=10
pair $\Rightarrow$ 2

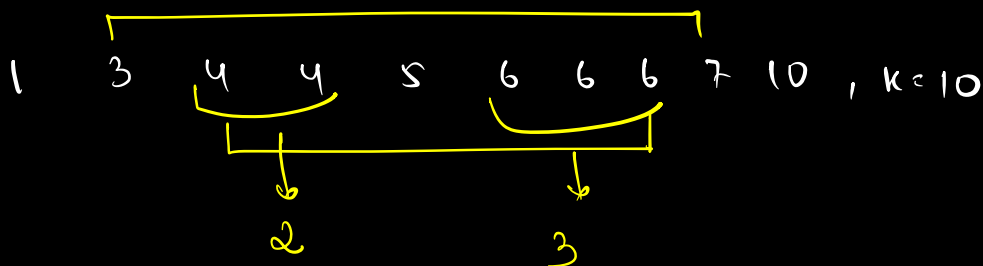
⑥

1 3 4 4 5 6 7 10 , k = 10

Diagram illustrating the merging of two sorted arrays [1, 3, 4, 4, 5, 6, 7] and [10] into a sorted array of size 10. The arrays are shown as segments of a line, with the second array starting at index 7. Brackets indicate the merging process, showing the final sorted array of size 10.

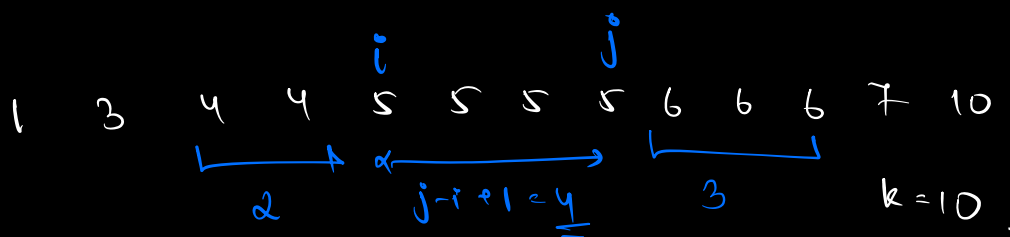
peris $\Rightarrow 1 + (2 \times 1) = 3$.

(11)



$$\text{pairs} \Rightarrow 1 + (2 \times 3) = 7 \text{ pairs}$$

(12)



$$\text{pairs} \Rightarrow 1 + (2+3) + 4C_2$$

$$= 1 + 6 + \frac{4!}{2!2!}$$

$$= 1 + 6 + 6$$

$$= 13 \text{ pairs}$$

① $\text{arr}[i] + \text{arr}[j] = k$

but no duplicates

comb $\rightarrow 1$

② $\text{arr}[i] + \text{arr}[j] = k$

and, n duplicates of i

m duplicates of j

comb. $n \times m$

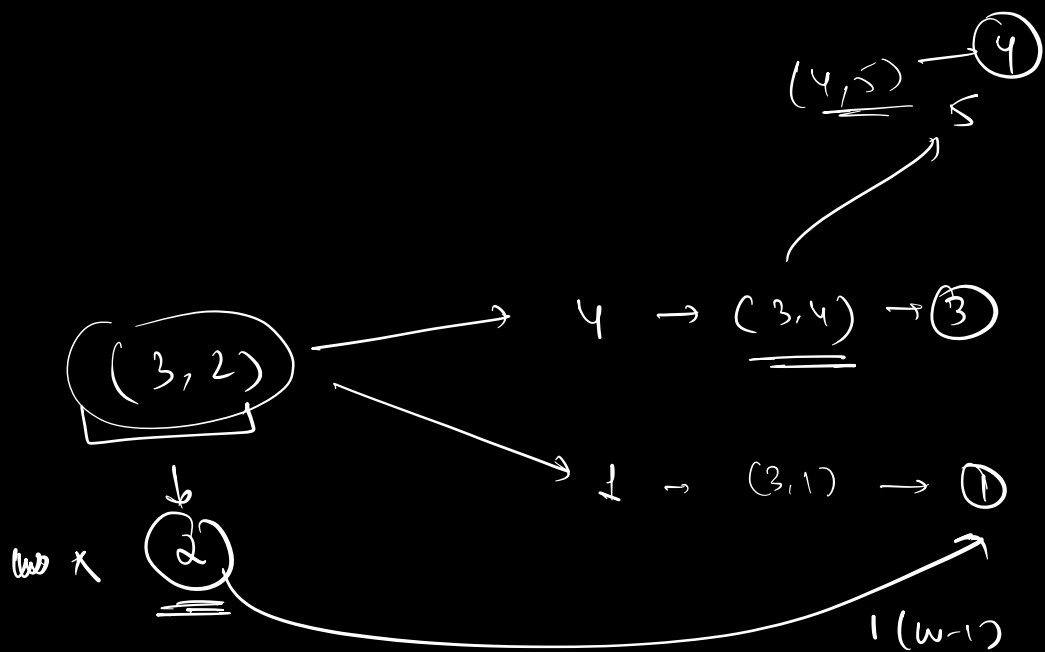
(11) $\text{arr}[i] + \text{arr}[j] = k$

and, $\text{arr}[i] = \text{arr}[j]$

$$\begin{aligned} T &= O(N) \\ S &= O(1) \end{aligned}$$

↓
elements $\Rightarrow N = j - i + 1$

$$\text{ways} = N_c = \frac{N!}{2!(N-2)!}$$



23
||
||
||
||